

DATA_AVAILABILITY_STATEMENT.md

Deliverable D-S2 — Data availability and ethics statement

Project: AMGCR Earthquake Research

Scope: California FDSN pilot (USGS events + EarthScope waveforms)

Date: 29 July 2026

This statement describes **where data come from, what is stored in the repository, what is excluded from version control, and ethical considerations**. It is based on [IRIS_DATASET_REPORT.md](#), [DATASET.md](#), [FINAL_DELIVERABLE_AUDIT.md](#), and [Research_Proposal_v1.md](#).

Related: [REPRODUCIBILITY_STATEMENT.md](#)

1. Data used in the California pilot

Role	Provider	Access	ObsPy client (as documented)
Earthquake catalog / hypocenters	U.S. Geological Survey (USGS) FDSN Event service	Public web service	<code>Client("https://earthquake.usgs.gov")</code>
Waveforms & station metadata	EarthScope Consortium (successor to IRIS Data Management Center)	Public FDSN datasetselect /station	<code>Client("EARTHSCOPE")</code>

Service URLs (repository documentation):

- USGS FDSN Event: <https://earthquake.usgs.gov>
- EarthScope data services: <https://www.earthscope.org/>
- EarthScope FDSN base (ObsPy): <https://service.earthscope.org>
- FDSN specification: <https://www.fdsn.org/webservices/>

Pilot query summary ([IRIS_DATASET_REPORT.md](#)): California bounding box (32.5°–42.0° N, 124.5°–114.0° W), 2024-01-01 to 2024-12-31, magnitude **4.0–7.5**; **one** **BHZ** MiniSEED trace per retained event, **300 s** from origin, **40 Hz** sampling for manifest-linked files.

Completed run (2026-07-29): 8 events in data/manifests/iris_california_pilot_events.csv; magnitudes **4.05–4.87**; stations **CI.ADO** (6 events) and **CI.USC** (2 events).

No credentials were required for this public pilot ([IRIS_DATASET_REPORT.md](#) §4).

2. Data formats and locations

Data type	Format	Location when present
Event manifest	CSV	data/manifests/iris_california_pilot_events.csv
Raw waveforms	MiniSEED (.mseed)	data/raw/iris/<event_id>/
Acquisition run summary	JSON	logs/iris_california_pilot_summary.json (after download)
Derived tables, metrics, NPZ, feature matrix	CSV, JSON, NPZ	reports/tables/, reports/eda/, reports/signal_analysis/, reports/preprocessing/, reports/features/
Figures	PNG (300 DPI)	reports/figures/

Rules: Raw MiniSEED should remain **immutable**; new pulls or parameter changes should use new downloads or documented manifests ([DATASET.md](#)).

3. Availability in the Git repository

3.1 Available without re-download (typical clone)

- **Event manifest** — data/manifests/iris_california_pilot_events.csv (authoritative list of the eight analysed events).
- **Derived analysis products** under reports/ (tables, JSON summaries, preprocessing stage NPZ, feature matrix) as committed in the project history.
- **Documentation** of query parameters and methods — docs/IRIS_DATASET_REPORT.md and phase reports.
- **Scripts** to re-fetch and recompute — scripts/download/, scripts/analysis/.

3.2 Not available in Git (intentionally excluded)

Per `.gitignore`:

Excluded	Pattern / path
Raw waveforms	<code>data/raw/</code>
Local logs	<code>logs/</code>
Generated figures	<code>figures/</code> (includes <code>reports/figures/</code>)

[FINAL_DELIVERABLE_AUDIT.md](#): a fresh clone does **not** contain MiniSEED files, acquisition JSON in `logs/`, or the **24** pilot PNGs unless they are regenerated or supplied in a **separate submission archive**.

3.3 Regaining excluded data

Need	Action
Raw MiniSEED + updated manifest (if re-run)	<code>python scripts/download/download_iris_california_pilot.py --event-count 8 --min-events 5</code>
Acquisition log	Produced by the same download script → <code>logs/iris_california_pilot_summary.json</code>
Figures	Re-run analysis scripts in order (REPRODUCIBILITY_STATEMENT.md §3–5)

Users may also obtain equivalent event and waveform data directly from **USGS** and **EarthScope** FDSN services using the parameters in [IRIS_DATASET_REPORT.md](#).

4. European and other data (context only)

The **programme's long-term focus** is European seismic networks ([PROJECT_CHARTER.md](#)). **No European waveform dataset is included** in this repository yet. Future work is expected to follow the same manifest-and-report pattern; **ORFEUS / EIDA** is cited as the intended archive class:

- ORFEUS Data Center / EIDA: <https://www.orfeus-eu.org/>

(European integration is **planned**, not part of the completed California pilot deliverable.)

Other documented sources (not used for California pilot results):

- Sample generic catalogue: `data/raw/catalogs/` (sample QuakeML).
- EarthESND Japan (K-NET) — optional reproduction track only ([DATASET.md](#), [PAPER_REPRODUCTION.md](#)).

5. Ethics and privacy

- **Human subjects:** The pilot uses **public seismological network data** only. No personal identifiers, survey responses, or human-subject research data are collected or processed.
- **Sensitive locations:** Event locations and station codes are **public catalog metadata** from USGS and EarthScope.
- **Licensing and attribution:** Data providers require **appropriate citation and use of FDSN services**; this project documents providers in reports and references (below). Users reusing data must comply with **USGS, EarthScope, and FDSN** terms applicable at the time of access.
- **GDPR:** The completed pilot does not process EU personal data; future European integrations may require network-specific policies ([RESULTS_AND_DISCUSSION.md](#) §9.3).

6. Limitations relevant to data use

Documented in project reports (not new claims):

- **Single component (BHZ), N = 8**, mixed magnitude types (mw, m1).
 - **Selection bias** toward station **CI.ADO** due to download policy ([IRIS_DATASET_REPORT.md](#), [EDA_REPORT.md](#)).
 - **Legacy MiniSEED** files may exist locally outside the current manifest (four files noted in EDA); **published analyses use eight manifest-linked files only**.
 - **Instrument response correction** was not applied in the completed preprocessing run (**0/8**); amplitudes in counts are **not** physical PGA/PGV without further processing ([PREPROCESSING_REPORT.md](#)).
 - California pilot data **do not** reproduce EarthESND paper geography or metrics ([IRIS_DATASET_REPORT.md](#) §5).
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7. Citations and references (data and software)

Data services and archives

U.S. Geological Survey. Earthquake Hazards Program — FDSN event web service. <https://earthquake.usgs.gov>

EarthScope Consortium. Data services. <https://www.earthscope.org/>

International Federation of Digital Seismograph Networks (FDSN). FDSN web services specification. <https://www.fdsn.org/webservices/>

ORFEUS Data Center / EIDA. European integrated waveform archives. <https://www.orfeus-eu.org/>

Software

Krischer, L., Megies, T., Barsch, R., Beyreuther, M., Lecocq, T., Caudron, C., and Wassermann, J. (2015). ObsPy: A bridge for earthquake science. *Seismological Research Letters*, 86(3), 765–771.

EEW context (literature cited in repository proposal)

Allen, R. M., and Kanamori, H. (2003). The potential for earthquake early warning in southern California. *Science*, 300(5620), 786–788.

Hoshiba, M., Iwakiri, K., Hayashimoto, N., Shimoyama, T., Hirano, K., Yamada, Y., Ishigaki, Y., and Kikuta, H. (2008). Outline of the 2007–2008 earthquake early warning experiments in Japan. *Earth Planets Space*, 60, 123–129.

Joshi, A., Singh, A. P., and Raman, B. (2026). EarthESND: Lightweight multiscale echo state network with dendritic neural network readout for earthquake early warning. *Computers and Electrical Engineering* (reference implementation only for this programme; see docs/EARTHESND_REVERSE_ENGINEERING.md).

Project documentation index

docs/PROJECT_CHARTER.md; docs/RESEARCH_DIRECTION.md;
docs/IRIS_DATASET_REPORT.md; docs/EDA_REPORT.md; docs/SIGNAL_ANALYSIS_REPORT.md;
docs/PREPROCESSING_REPORT.md; docs/FEATURE_ENGINEERING_REPORT.md;
docs/RESULTS_AND_DISCUSSION.md; reports/Research_Proposal_v1.md §12.

Document control: D-S2 data availability statement v1.0 — 29 July 2026. No source code modified to produce this file.